EE1102 : Introduction to Programming
Aug-Nov 2023 - Assignment 1

Things to Know for this Assignment

All writing and reading requires including stdio.h:

```
#include<stdio.h>   /* this is a comment line (can be on separate line too) */
```

Basic structure of a C program:

```
#include….
int main(){     /* start of the main function. All execution starts here */
        ….      /* code */
        return (...)     /* return value to the operating system */
}               /* end of the main program */
```

To write something, use printf("...").

```
printf("This is a line\n");
This is a line
```

To print multi-line output

```
printf("This is\na line\n");  /* each \n starts a new line */
This is
a line
```

To declare an integer variable and assign it a value

```
int a;
a=4;
```

To write something that includes an integer variable (place holder for integer is %d)

```
printf("The value of a = %d\n",a); /* output printed, with %d replaced by value */
The value of a = 4
```

Addition of variables and constants

```
Int b=10;
Int c=15;   /* can give the variable a value inside the declaration itself */
Int d,e;
d=a+b+c;  /* addition of variables */
e=a+4+c;  /* addition of variables and constants (here 4 is a constant) */
```

New additions to this section:

To declare a string variable, and assign it a value

```
char name[ ] = ”Abc Xyz”;
```

The syntax will be explained later, but this is a simple way to declare string constants. To print some strings, use

```
printf(“My name is %s, and my home is in %s\n”, name, city);
```

%s is the symbol used to indicate a string. It will be taken from the list of variables following the format string.

To declare a real number, and assign it a value

```
float a = 4.5;
```

This creates a real number that uses 32 bits of storage. The mantissa is 24 bits in size including sign, and the exponent is 8 bits in size. Note that exponent cannot exceed 7 bits as there is an offset. To print float numbers use

```
printf(“I am %f cm tall\n”, 66.5);
```

Note that the value for height can also be taken from a constant

To declare a long int variable. This is a little strange. “int” is not a fixed size. It depends on the cpu of the computer. So a long integer variable is called “long long int”

```
long long int aa = 123456789;
printf("aa = %lld\n", aa);
```

This creates an integer that uses 64 bits of storage. Such an integer can store much larger size integer values. For example,

```
long long int amnt=1000;    /* amount in crores to be printed out */
Int crore = 10000000;       /* Number of Rs in 1 crore */
amnt = amnt*crore;          /* amnt converted to Rs. */
printf("1000 crores is %lld Rs.", amt );
```

To declare a long real number, and assign it a value

```
double b = 3.5e250
```

This creates large real numbers whose mantissa or exponent will not fit into a float variable. For instance if I want to store pi as a constant. We use

```
double pi = 3.141592653589793
```

This number is well within the size of numbers that can be held by a float. But the number of mantissa digits are too large for float. We need a 60 bit float, which is called double.

Assignment

Problem 1. Write a C program called diff.c that prints the difference of 2 integers in the format shown in the example below. Use variables of type int. Try changing the values for each variable and check if the output is correct.

Example: If the 2 integers you want to use are 10 and 25, the program

should print

10 - 25 = -15

Problem 2. Write a C program called print_my_details.c that prints your roll number, a comma, and your name in the first line, and your hostel and room number in the second line. Note that each of the items is in a separate variable.

Example: If a student's roll number is EE20B888 and name is Abc Xyz, and he is in Mandakini Hostel in Room 42, the output printed when we run the program should be

char rollNo[ ] = "EE20B888"; char name[ ] = "Abc Xyz";

char hostel[ ] = "Mandikini"; int room = 42;

EE20B888, Abc Xyz
Room 42, Mandakini Hostel

Problem 3. Write a C program that takes three numbers and calculates the ratio of the sum of the squares of the numbers divided by the sum of the numbers in the format shown in the example below. Use variables of type float .. Try changing the values for each variable and check if the output is correct.

Example: If the 3 numbers you want to use are 10.5, 25.8, and -5.3, it should print

A = 10.5; B = 25.8; C = -5.3;

(A^2 + B^2 + C^2) / (A + B + C) = 803.98

Problem 4. How big an integer can be stored in an int variable? This requires you to calculate how large 2^31 can be. Note that it is 2^31 and not 2^32 since one bit is used for sign.

Problem 5. Assume that the Institute received 600 crores (1 crore is 10 million Rupees) as its budget this year. Assume that a lab computer costs 40,000 Rupees. How many computers could this amount buy? Print out the answer using

The budget is Rs. ………. . We can buy ………….. computers.

where the numbers should be stored as integer or long long integer depending on how large it is. Justify your choice of formats used. Even though you could use long long int for both, I want you to use the correct format for each number.